ROSP-TR2 Lab
Simulating a Fiber-optic Communication System

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Abstract

This is a guide for a simulation project on implementing a basic fiber-optic communication system. The project consists of implementing a transmitter, an optical fiber channel using the split-step Fourier method and a receiver. The bit error rate is plotted as a function of the signal-to-noise ratio for several constellations in a wavelength-division multiplexing system.

Contents

1 Introduction

We simulate a basic optical fiber communication system in this project. We consider a simplified system that can be implemented in a short course.
The project consists of two parts. In the first part in Section 2, you will design and evaluate a transmitter (TX) and receiver (RX) for the dispersive optical fiber (namely, in the absence of nonlinearity). In the second part in Section 3, you will implement the split-step Fourier method (SSFM) and design a wavelength-division multiplexed system for communication over the nonlinear dispersive optical fiber.

Deliverable. You will submit your code and a report in which you answer the questions in this guide. In particular, you will present the final plots on the bit error rate (BER) versus signal-to-noise ratio (SNR) in singlet-channel transmission and WDM, in different system configurations specified in this guide. The report can be written in LaTeX. The grading is based on the results, code, report, performance in the labs, and a final discussion if required.
The students start working on the project in the labs, and complete it at home during the semester. The theoretical basis for the project can be found in DCOM slides [1].
The software is preferably written in Python, but Julia or MATLAB may also be used[1]. The code must be written by the students. The code generated by AI or existing code will not be accepted. There are toolboxes that implement the building blocks in this project. The use of these toolboxes is not permitted. The instructor will provide office hours and hands-on help.

1.1 Overview of the simulated communication system

The block diagram. The *separation theorem* for point-to-point data communication implies that the source and channel can be layered separately as shown in Fig. 1 [2]. The data (*e.g.*, a camera image) is initially in the analogue domain. The analogue signal is converted to a sequence of bits by the source encoder (consisting of sampling, quantization and data compression). This digital representation has a number desirable advantages [2, Chap. 4.1.1]. After applying digital signal processing, the bits sequence is transformed back to a continuous-time signal by the channel encoder.
The channel encoder consists of channel coding, modulation (bits-to-symbols and symbols-to-signal mapping) and baseband-to-passband shift. The inverse of these layers is the channel decoder. In this lab, we only implement the part from the input of the channel encoder to the output of the channel decoder. Furthermore, we ignore channel error-correction coding and consider a baseband model.
The foundations of the digital communication are reviewed in [2–4].

[1] MATLAB is used for demonstration in this guide due to its compact syntax. A free open source implementation of MATLAB is GNU Octave

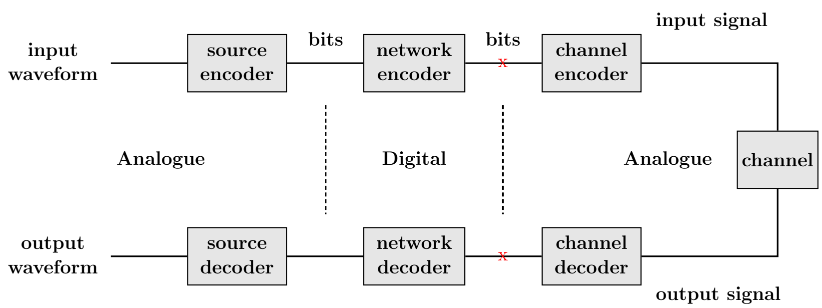


Fig. 1: Block diagram of a communication system.

Setting up a simulation environment. Sign in to your computer account, choosing a French keyboard for the password. Create a folder "tr2-lab-names," where "names" is the last name of the group members. You will save all files here.
Create a simulation environment as follows.

a. Matlab:

   (a) Run Matlab 2025, located in section Development

   (b) Create the following Matlab files (m-files): parameters.m, main.m, source.m, mod.m, mux.m, channel-linear.m, channel.m, equalizer.m, demux.m, demod.m, detector.m and ber.m.

b. Python:

   (a) If you use desktop, set up a local integrated development environment, such as VSCode.
   If you you use command line, SSH into a school server, issuing `ssh node-name -J username@ssh.enst.fr`, where `node-name` is the name of the server, *e.g.*, `lame21`.

   (b) Create a Python virtual environment `python -m venv tr2-lab-name`, activate it, `source bin/activate` and install libraries `pip install numpy scipy matplotlib`.

   (c) Create the files mentioned above. They are organized in modules. For instance, you may create a module `parameters.py` that contains all static objects and constants, and a module `propagation.py` that contains all functions required to simulate the fiber. You import them in `main.py` and other scripts.

1.2 Channel model.

The stochastic NLS equation. Let $Q(\tau, \ell) : \mathbb{R} \times \mathbb{R} \mapsto \mathbb{C}$ be the complex envelope of the propagating signal as a function of time τ and distance ℓ. Pulse

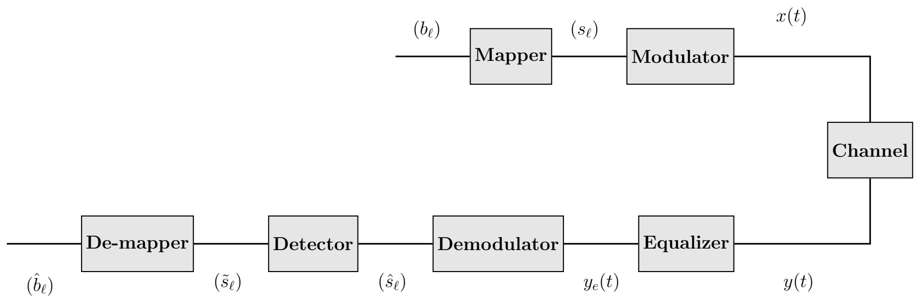


Fig. 2: Simplified block diagram.

propagation in optical fiber with ideal distributed amplification is modeled by the stochastic nonlinear Schrödinger (NLS) equation [5, Eq. 7]

$$\frac{\partial Q(\tau,\ell)}{\partial \ell} = -\underbrace{\frac{j\beta_2}{2}\frac{\partial^2 Q(\tau,\ell)}{\partial \tau^2}}_{\text{dispersion}} + \underbrace{j\gamma|Q(\tau,\ell)|^2 Q(\tau,\ell)}_{\text{nonlinearity}} + \underbrace{N(\tau,\ell)}_{\text{noise}}, \tag{1}$$

where β_2 is the chromatic dispersion coefficient, γ is the nonlinearity parameter and $j = \sqrt{-1}$. Further, $N(\tau,\ell)$ is bandlimited circular symmetric complex Gaussian noise with autocorrelation

$$\mathsf{E}\left\{N(\tau,\ell)N^*(\tau',\ell')\right\} = \sigma_0^2\delta_B(\tau-\tau')\delta(\ell-\ell'),$$

where $\delta(x)$ is the Dirac Delta function, $\delta_B(x) = B\,\mathrm{sinc}(Bx)$, $\mathrm{sinc}(x) = \sin(\pi x)/(\pi x)$, B is the noise bandwidth, σ_0^2 is the in-band noise power spectral density (PSD) and E is the expected value. The transmitter is located at $\ell = 0$ and the receiver at $\ell = \mathcal{L}$.

The stochastic NLS equation (1) models the chromatic dispersion, Kerr nonlinearity and the amplified spontaneous emission (ASE) noise. It is assumed that the fiber loss is perfectly compensated by the distributed amplification.

Constraints. The transmitter is subject to the average power constraint

$$\lim_{T\to\infty}\frac{1}{T}\mathsf{E}\left\{\int_{-\frac{T}{2}}^{\frac{T}{2}} |Q(\tau,0)|^2 \mathrm{d}\tau\right\} \leqslant \mathcal{P}. \tag{2}$$

It is assumed that $Q(\tau,\ell)$ is bandlimited to B Hz for all ℓ, $0 \leqslant \ell \leqslant \mathcal{L}$.

Question 1. The variables Q, τ and ℓ are measured in $\sqrt{\text{Watt}}$ ($\sqrt{\text{W}}$), second (s) and meter (or kilometer), respectively. Find out the units of β_2, γ and σ_0^2. □

Normalization. We will find it convenient to normalize the NLS equation.

Question 2. Consider the change of variables

$$q = \frac{Q}{\sqrt{P_0}}, \qquad t = \frac{\tau}{T_0}, \qquad z = \frac{l}{\mathcal{L}}.$$

Show that if we choose

$$P_0 = \frac{2}{\gamma\mathcal{L}}, \qquad T_0 = \sqrt{\frac{|\beta_2|\mathcal{L}}{2}}, \qquad L = \mathcal{L}, \tag{3}$$

we obtain from (1)

$$\boxed{j\frac{\partial q(t,z)}{\partial z} = \frac{\partial^2 q(t,z)}{\partial t^2} + 2|q(t,z)|^2 q(t,z) + n(t,z),} \tag{4}$$

where $0 \leqslant z \leqslant 1$, and we assumed $\beta_2 < 0$. Here, $n(t,z)$ is circular symmetric complex Gaussian noise with autocorrelation

$$\mathsf{E}\left\{n(t,z)n^*(t',z')\right\} = \sigma^2 \delta_{B_n}(t-t')\delta(z-z'), \tag{5}$$

in which

$$\sigma^2 = \frac{\sigma_0^2 \mathcal{L}}{P_0 T_0}, \qquad B_n = BT_0. \tag{6}$$

□

In what follows, we consider the normalized NLS equation (4).

1.3 Signal and system parameters

We consider a fiber with parameters in Tab. 1, $\mathcal{L} = 1000$ km and $B = 1$ to 10 GHz. We assume a quadrature amplitude modulation (QAM) constellation $\mathcal{C}$ with $M = 16$ points; see Fig. 3(b).

Question 3. Enter the constants in Tab. 1 in `parameters.m`
Convert the dispersion parameter D with unit ps/(nm − km) to dispersion coefficient β_2. Recall that $\beta_2 = -\frac{\lambda_0^2}{2\pi c}D$, where $c = 3 \times 10^8$ m/s is the speed of light.
Calculate the scale parameters in (3).
Convert the loss coefficient α from unit dB/km to m^{-1}, using

$$\alpha = 10^{-4}\log(10) \times a_{\mathrm{dB}}, \quad \mathrm{m}^{-1}.$$

See Appendix A for change of unit.
Calculate the noise PSD $\sigma_0^2 = n_{sp} h\alpha f_0$, where f_0 is the carrier frequency, and its normalized value σ^2.
The constants, scale factors, physical and normalized variables should be in `parameters.m`.

□

Question 4. Consider the average power $\mathcal{P} = 6$ mW. Compute 16 symbols in the constellation $\mathcal{C}$ such that $\mathsf{E}|s_k|^2 = \mathcal{P}$. Plot the constellation $\mathcal{C}$.

Table 1: Fiber and Noise Parameters

a_{dB}	0.2 dB/km	fiber loss
D	17 ps/(nm-km)	chromatic dispersion
γ	1.27 $\mathrm{W}^{-1}\mathrm{km}^{-1}$	nonlinearity parameter
n_{sp}	1	spontaneous emission factor
h	$6.626 \times 10^{-34} \mathrm{J} \cdot \mathrm{s}$	Planck's constant
λ_0	1.55 μm	carrier wavelength

2 Communication over dispersive channel

Consider first consider the NLS equation in the absence of nonlinearity

$$j\partial_z q(t,z) = \partial_{tt} q(t,z) + n(t,z). \tag{7}$$

2.1 Transmitter

Binary source. A discrete source is a discrete-time stochastic process.

Question 5. Consider a Bernoulli stochastic process, *i.e.*, a sequence

$$b^N = (b_0, b_1, \cdots, b_{N-1}),$$

where b_i are random variables drawn independently from a Bernoulli probability distribution function (PDF)

$$p(b_i) = \begin{cases} 1-p, & b_i = 0, \\ p, & b_i = 1. \end{cases}$$

Write an m-function `b = source(N, p)` that generates b^N, and test it with $N = 1024$ and $p = 1/2$. You have access to a uniform random variable $U \sim [0, 1]$.

□

Question 6. In practice, the input bit stream is a maximum length pseudo-random binary sequence (PRBS). This is a periodic ± 1 sequence with a specific auto-correlation function, generated by linear feedback shift registers. Upgrade the source function and implement a PRBS source.

□

Modulation. The modulation consists of bits-to-symbols and symbols-to-signal mapping.

Question 7. *Bits-to-symbols mapping.* There are M symbol points in the constellation. Map each symbol point to a binary sequence of length $\log M$. Can you present a mapping such that the Hamming distance between the binary sub-sequences associated with the neighbor symbol points is 1? That is the Gray mapping.

Divide b^N into $n = N/\log M$ sub-sequences, each of which with $\log(M)$ bits. Map each sub-sequence into a symbol in the constellation and obtain a sequence of complex numbers $s^n = (s_0, s_1, \cdots, s_{n-1})$, where $s_i \in \mathcal{C}$ are *symbols.*
Write an m-function `s = bit_to_symb(b, cnt)` where cnt is the constellation (vector or matrix of all symbol points).

$\square$

It can be shown that if $p = 1/2$ then s^n is a sequence of complex random variables drawn i.i.d. from the uniform distribution over $\mathcal{C}$.

Symbols-to-signal mapping. Linear modulation consists of expanding a signal $q(t, 0)$ onto an orthonormal basis. The Nyquist-Shannon sampling theorem proved below provides a suitable representation.

Question 8. Consider the vector space U of the finite-energy signals $\mathbb{R} \mapsto \mathbb{R}$ band-limited to B Hz with the inner product $\langle f, g\rangle = \int fg$, $f, g \in U$. Prove that $\left\{\operatorname{sinc}(Bt-\ell)\right\}_{-\infty}^{\infty}$ is a unit-power orthogonal basis for U. Compute $\langle f, \operatorname{sinc}(Bt-\ell)\rangle$, where $f \in U$.

$\square$

Thus, we have

$$\begin{aligned} q(t, 0) &= \sum_{\ell=\ell_1}^{\ell_2} s_\ell p(t - \ell T_s) \\ &= \sum_{\ell=\ell_1}^{\ell_2} s_\ell \operatorname{sinc}(Bt - \ell), \end{aligned} \tag{8}$$

where $(s_k)_{k\in\mathbb{Z}}$ is a sequence of zero-mean independent identically distributed (i.i.d.) random variables drawn uniformly from $\mathcal{C}$, $p(t) = \operatorname{sinc}(Bt)$, $T_s = 1/B$, $\ell_1 = -\lfloor n_s/2\rfloor$, $\ell_2 = \lceil n_s/2\rceil - 1$, and $\mathsf{E}|s_k|^2 = \mathcal{P}$.
Define the Fourier transform of $f \in L^1(\mathbb{R} \mapsto \mathbb{R})$ with convention:

$$\hat{f}(\omega) = \mathcal{F}(f)(\omega) = \int_{-\infty}^{\infty} f(t)e^{-j\omega t}\mathrm{d}t. \tag{9}$$

Question 9. The Fourier transform of the pulse shape $\hat{p}(\omega) = \mathcal{F}(p)(\omega)$ is a rectangle if $p(t) = \operatorname{sinc}(Bt)$. In practice, a smooth version of this rectangle called root raised-cosine (RRC) filter is used.
Write a function `rrc(t, B, r)` for the RRC filter in time, where $r \in (0, 1]$ is the excess bandwidth factor. You can use the expression in time or frequency [2, Sec. 6.3.1, Eq. 6.17, 6.18], [3, Eq. 11.29, Eq. 11.31]. In this project, write your code for $p(t) = \operatorname{sinc}(Bt)$; once the software works, switch $p(t)$ to the RRC filter with $r = 0.3$.

$\square$

Question 10. The project has two m-functions main.m (one for this Section and one for Section 3), in which you set up and run the project. You will gradually complete this file.
Complete the following main.m file.

```
% time mesh
T  = 400; % you have to choose this
N  = 2^10; % you have to choose this
dt = T/N;
t  = ... % should be from -T/2 to T/2 with length N

% frequency mesh
F  = 1/dt;
df = 1/T;
f  = ... % should be from -F/2 to F/2 with length N

% bits to signal
M = 16; % size of the constellation
n = 3; % number of symbols (or sinc functions); change this later
nb = ... ; % number of bits
p =1/2; % probability of zero
b = source(nb, p); % Bernoulli source, random bits sequence
s = bit_to_symb(b, cnt); % symbol sequence; could be inside modulator.m
q0 = modulator(t, s, B); % transmitted signal
```

Write an m-function for modulator `mod(t, s, B)`, implementing (8).

```
function q0t = mod(t, s, B)

Ns = len(s)  % number of symbols
l1 = ...
l2 = ...

q0t = ...

end
```

Plot $q(t, 0)$ in time and in frequency. Make sure that the basis elements have unit energy: $\|\mathrm{sinc}(x)\|^2_{L^2(\mathbb{R})} = \left\|\sqrt{B}\,\mathrm{sinc}(Bx)\right\|^2_{L^2(\mathbb{R})} = 1$.

□

Remark 1. The code in this guide should be viewed as pseudo-code. You have to complete the code, add missing variables, arguments, etc. You may have to change the parameters also in order to obtain meaningful results, especially in places where this is indicated. □

Question 11. The time interval between two consecutive sinc functions is $1/B$. Thus, the symbol rate is $R_s = 1/B$ symbols/s. Express the bit rate R_b in bits/s in terms of R_s.

□

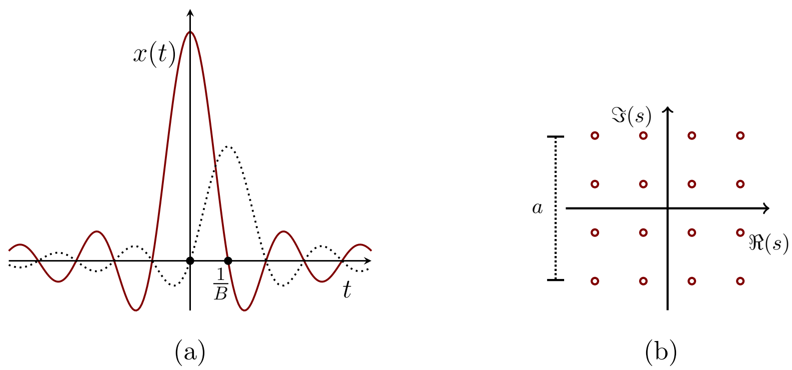


Fig. 3: (a) Modulation using sinc functions; (b) a 16-QAM constellation.

2.2 Dispersive optical fiber

Continuous-time model. Let us first simplify the partial differential equation channel (7) to a more familiar form.

Question 12. Define the Fourier transform of $q(t, z)$ with respect to t for fixed z as $\hat{q}(\omega, z) = \mathcal{F}(q(t, z))(\omega)$. We solved (7) with $n(t, z) = 0$ in the class. Solve (7) in the presence of noise and show that it is equivalent to the *additive white Gaussian noise (AWGN) channel*

$$\hat{q}(\omega, z) = \hat{h}(\omega, z)\hat{q}(\omega, 0) + \hat{z}(\omega), \tag{10}$$

where $\hat{q}(\omega, z)$ is the Fourier transform of $q(t, z)$. Determine the channel transfer function $\hat{q}(\omega, z)$. Characterize the stochastic process $\hat{z}(\omega)$, *i.e.*, determine the PDF of $\hat{z}(\omega)$, its mean $\mathsf{E}\hat{z}(\omega)$ and the correlation function $\mathsf{E}(\hat{z}(\omega)\hat{z}^*(\omega'))$.
In the time domain we have

$$q(t, z) = h(t, z) * q(t, 0) + z(t), \tag{11}$$

where $*$ is convolution. Determine the channel impulse response $h(t, z)$. Determine the PDF of $z(t)$, and its mean and correlation.

□

Discrete-time model. The continuous-time model (7), (11) or (10) should be discretized to a discrete-time model for simulation. To formulate a discrete problem, we need to discretize time and space, thereby the channel and the constraints.
The signal $q(t, z)$ can be sampled in time at the Nyquist rate $1/B$ to obtain a vector

$$\mathbf{q}(z) = \begin{pmatrix} q_1(z) & q_2(z) & \ldots & q_N(z) \end{pmatrix}.$$

However, to discretize the integral that underlies the convolution in (11) (or the time derivative in (7)), we sample the signal at a sufficiently small simulation step size dt, which can be much less than the Nyquist rate.

Question 13. Discretize the signal power (2), by discretizing the integral into a Riemann sum. □

Question 14. The channel can be discretized in the frequency domain by sampling (10) on a frequency mesh obtained from the time mesh.
Discretize the Fourier integrals

$$\hat{x}(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}\mathrm{d}t, \quad x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \hat{x}(\omega)e^{j\omega t}\mathrm{d}\omega,$$

on the time-frequency mesh introduced above in the main.m file, and express these Fourier integrals in terms of the discrete Fourier transforms (DFTs)

$$\hat{x}_k = \sum_{n=0}^{N-1} x_n e^{-j\frac{2\pi nk}{N}t}, \quad x_n = \frac{1}{N}\sum_{n=0}^{N-1} \hat{x}_k e^{j\frac{2\pi nk}{N}t}.$$

Use this result to approximate $\hat{q}(\omega, z)$ in terms of $\hat{\mathbf{q}}(z) = \mathrm{DFT}(\mathbf{q}(z))$.
To summarize, write down the channel model in the frequency $\hat{\mathbf{q}}(0) \mapsto \hat{\mathbf{q}}(z)$ and in time $\mathbf{q}(0) \mapsto \mathbf{q}(z)$, and their input constraints.

□

2.3 Receiver

Equalization. Equalization is the task of inverting the channel transformation in the absence of noise. For the dispersive channel, this can be done in the frequency domain

$$\begin{aligned}\hat{q}_e(\omega, \mathcal{L}) &\triangleq \hat{h}^{-1}(\omega, \mathcal{L})\hat{q}(\omega, \mathcal{L}) \\ &= \hat{q}(\omega, 0) + \hat{h}^{-1}(\omega, \mathcal{L})\hat{z}(\omega) \\ &\triangleq \hat{q}(\omega, 0) + \hat{w}(\omega),\end{aligned}$$

where $\hat{w}(\omega) = \hat{h}^{-1}(\omega, \mathcal{L})\hat{z}(\omega)$. In the time domain

$$q_e(t, \mathcal{L}) = q(t, 0) + w(t), \tag{12}$$

where, *e.g.*, $q_e(t, \mathcal{L}) = \mathcal{F}^{-1}(\hat{q}_e(\omega, \mathcal{L}))$.

Question 15. Characterize the stochastic processes $w(t)$ and $\hat{w}(\omega)$, namely, state their PDFs and the statistical properties (the mean and the autocorrelation function).

□

Question 16. Write an m-function `equalize(t, qzt, z)` that outputs $q(t, 0)$ given $q(t, z)$ in the absence of noise.

```
function [qzte, qzfe] = equalize(t, qzt, z)

f = ...         % get the f vector from t vector
```

```
qzf = ...      % input in frequency
qzfe = qzf.* ... % output in frequency


qzte = ... % back to time

end
```

□

Question 17. Write an m-function `channel(t, q0t, beta2, z, sigma2, B)` that outputs a realization of the stochastic process $q(t,z)$ given $q(t,0)$. Write these functions for any values for parameters (not just, *e.g.*, for $\beta_2 = -2$).
You need to discretize noise $n(t,z)$ to a vector $\mathbf{n}(z)$, and (5) to a discrete autocorrelation function $\mathsf{E}(\mathbf{n}(z)\mathbf{n}^H(z))$. Hint: equate the total noise power in the continuous and discrete models.

```
function [qzt, qzf] = channel(t, q0t, beta2, z, sigma2, B)

f = ...        % get the f vector from t vector

q0f = ...      % input in frequency
qzf = q0f.* ... % output in frequency

% add noise correctly. In time or frequency? In what bandwidth?
qzf = qzf + ...

qzt = ... % back to time

end
```

Write a similar m-function `equalize(t, qzt, beta2, z)` that outputs $q(t,0)$ given $q(t,z)$ in the absence of noise.
Add the following to your main.m file.

```
A   = 1;
q0t = A*exp(-t.^2); % Gaussian input, for testing
q0f = ... % input in frequency
% plot below the input in t & f. Coarse-tune T and N.

% propagation
beta2 = -2; % choose alll signal and system parameters
[qzt, qzf] = channel(t, q0t, beta2, z, sigma2, B); % output in t,f
plot(t, q0t, ...) % plot input output in t & f; tune T and N accordingly

% equalization and comparison
[qzte, qzfe] = equalize(t, qzt, beta2, z);    % equalized output
plot(t, q0t, ...) % plot input & equalized output in t
compare(q0t, qzte) % you should write the function compare
```

$\square$

Demodulation. The demodulation consists of obtaining the received symbols at the output via projection

$$\begin{aligned}\hat{s}_\ell &= \frac{\langle q_e(t,\mathcal{L}), \operatorname{sinc}(Bt-\ell)\rangle}{\langle \operatorname{sinc}(Bt-\ell), \operatorname{sinc}(Bt-\ell)\rangle} \\ &= B\int_{-\infty}^{\infty} q_e(t,\mathcal{L})\operatorname{sinc}(Bt-\ell)\mathrm{d}t. \end{aligned} \tag{13}$$

Question 18. Write an m-function `shat = demod(t, qzte, B)` to recover $\hat{s}^n$ from $q_e(t,\mathcal{L})$, by implementing (13). Make sure that the basis elements have unit energy: $\|\operatorname{sinc}(x)\|^2_{L^2(\mathbb{R})} = \left\|\sqrt{B}\operatorname{sinc}(Bx)\right\|^2_{L^2(\mathbb{R})} = 1$.

```
function shat = demod(t, qzte, B, Ns)

shat = ...

end
```

$\square$

Detection. The optimal receiver is the *maximum likelihood (ML) detection* applied to the joint conditional PDF $p(\hat{s}^n|s^n)$, *i.e.*,

$$\tilde{s}^n = \underset{s^n\in\mathcal{C}^n}{\arg\max}\quad p(\hat{s}^n|s^n). \tag{14}$$

Question 19. Substitute (12) into (13) and obtain a discrete-time channel $s^n \mapsto \hat{s}^n$. Show that the channel $s^n \mapsto \hat{s}^n$ is memoryless, *i.e.*,

$$p(\hat{s}^n|s^n) = \prod_{\ell=1}^{n} p(\hat{s}_\ell|s_\ell). \tag{15}$$

Therefore the maximization in (14) can be performed separately per component $p(\hat{s}_\ell|s_\ell)$. Write the conditional PDF $p(\hat{s}_\ell|s_\ell)$, apply the ML detection (14), find decision regions and prescribe the optimal receiver.
Write an m-function `stilde = detector(shat, cnt)` that implements the ML detection. The function takes the received-equalized symbols $\hat{s}^n$ and outputs the ML estimates $\tilde{s}^n$.

```
function stilde = detector(shat, cnt)

stilde = ...

end
```

□

Demapping. This step is symbols-to-bits mapping, that you should implement in `bhat = symb_to_bit(stilde, cnt)` where cnt is the constellation. Use a look-up table for mapping and demapping.

Question 20. Check first the end-to-end implementation when noise is zero. In this case, you should obtain $\hat{\mathbf{s}} \approx \mathbf{s}$ with high accuracy.
Complete the main.m file.

```
% modulation
nb = 64; % number of bits
M = 16; % order of modulation
n = ... % number of symbols
b= ... % random bit sequence
s = bit_to_symb(b, cnt);
q0 = mod(t, s, B);

% propagation & equalization. Set the noise to zero for now

% demodulation
shat = demod(t, qzte, B);
shat = detector(shat, cnt);
bhat = symb_to_bit(shat, cnt);
```

Do you obtain $\hat{\mathbf{s}} = \mathbf{s}$ and $\hat{\mathbf{b}} = \mathbf{b}$ in the absence of noise?

□

Inter-symbol interference. One type of distortion in communication is *inter-symbol interference (ISI)*, explained in the class.

Question 21. Simulate the output for the input

$$q(t,0) = A_1 e^{-\frac{(t+t_0)^2}{2D^2}} + A_2 e^{-\frac{(t-t_0)^2}{2D^2}},$$

where $A_1 = 1$, $A_2 = 2$, $D = 1$ and $t_0 = 4$. Plot $q(t,0)$ and $q(t,1)$ (without equalization) in the same graph.
Does dispersion give rise to ISI in time (*i.e.*, the two parts in $q(t,0)$ that are non-overlapping at $z = 0$ overlap at $z = 1$)? Repeat for $t_0 = 7$. Is ISI completely canceled with equalization?
Does dispersion give rise to interaction between frequency components? Which domain is preferred for modulation?

□

2.4 Performance evaluation

In this section, we compute the BER as a function of SNR, numerically and analytically.

Back-to-back setup. Make sure that you obtain $\hat{s}^n = s^n$ and $\hat{b}^n = b^n$ in the back-to-back setup where the channel is $q(t, \mathcal{L}) = q(t, 0)$. Also, make sure that you obtain zero error in the noise-free channel with equalization.

Noisy channel. We now turn to the noisy channel.

Question 22. *(Simulated BER).* Add noise to the channel and compute the bit error rate $\text{BER} = p(\hat{b}_i \neq b_i)$ (or the symbol error rate). Complete the main.m file.

```
a = 1; % constellation radius
M = 2; % the number of points in the constellation
ser = ser(s,shat); % symbol error rate
ber = ber(b, bhat); % bit error rate

% plot the constellation at TX and RX. Change the annotation
plot(real(cnt), imag(cnt), '*', real(shat), imag(shat), '.');
% plot the BER
plot(snr, ber)
```

The BER should be plotted as a function of the $\text{SNR} = \mathcal{P}/P_n$ at TX, where $\mathcal{P}$ is the signal power and P_n is the total noise power (from $z = 0$ to $z = \mathcal{L}$). To compute $\mathcal{P}$, substitute (8) into (2) and prove that the signal power equals to the constellation power, *i.e.*, $\mathcal{P} = \mathsf{E}|s_k|^2$, $\forall k$. When $p = 1/2$,

$$\mathsf{E}|s_k|^2 = \frac{1}{M} \sum_{s \in \mathcal{C}} |s|^2, \tag{16}$$

which can be computed easily for a given constellation. The SNR can thus be scaled by varying a.

You can choose the number of symbols $n = 10000$, run the code $N_r = 1$ time, and count the errors. It is, however, easier to choose, say, five symbols $n = 5$, and run the same code $N_r = 500$ times in parallel over 4 CPU cores. You can change n and N_r as long as the the graphs are insensitive to the values of these parameters.

The final results of this section are: (i) constellation plot; (ii) the plot of the BER or SER as a function of SNR, for an M-QAM constellation with $M = 2, 4, 8, 16$ in the same figure.

□

Question 23. *(Analytic BER.)* Let $\mathcal{C}$ be an M-QAM constellation with radius a (see Fig. 3 (b)). Compute the average power of $\mathcal{C}$ in terms of M and a by evaluating (16).

Optional. Compute the BER as a function of SNR analytically. Hint: You already found $p(\hat{s}_\ell | s_\ell)$ analytically, and prescribed the operation of the ML receiver. Determine the decision regions of the ML, write down the error event E and compute $\text{SER} = p(E)$. How is BER related to the SER?

□

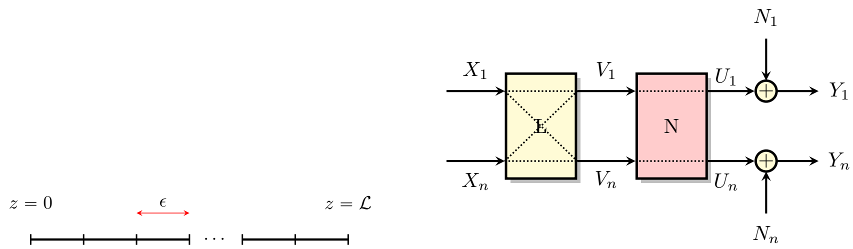


Fig. 4: Split-step Fourier method.

3 Communication over nonlinear dispersive channel

We now consider the NLS equation with general dispersion, nonlinearity and noise.

$$\begin{aligned}\frac{\partial q(t,z)}{\partial z} &= j\beta_0 q(t,z) + \beta_1 \frac{\partial q(t,z)}{\partial t} - \frac{j\beta_2}{2}\frac{\partial^2 q(t,z)}{\partial t^2} - \frac{\beta_3}{3!}\frac{\partial^3 q(t,z)}{\partial t^3} + \cdots \\ &+ j\gamma |q(t,z)|^2 q(t,z) + n(t,z), \end{aligned} \tag{17}$$

where β_k is the k-th order dispersion coefficient. We use the notation $q(t,z)$ instead of $Q(\tau,\ell)$ that is better for coding.

3.1 Split-step Fourier method

We solve (17) numerically using a basic version of the SSFM. Discretize the channel in z in intervals of length ϵ. In each segment, perform a linear, nonlinear and noise step.

Chromatic dispersion. This is a generalization of the linear step that you performed earlier.

Question 24. Show that the solution of the linear noiseless part of (17) is $\hat{q}(\omega,z) = e^{j\beta(\omega)z}\hat{q}(\omega,0)$, where $\beta(\omega) = \beta_0 + \beta_1\omega + \frac{1}{2}\beta_2\omega^2 + \cdots$. □

The dispersion step is thus $\hat{q}(\omega,\epsilon) = e^{j\epsilon\beta(\omega)}\hat{q}(\omega,0)$.

Kerr nonlinearity. Recall that the solution of the nonlinear part $\partial_z q = j\gamma|q|^2 q$ of (17) is

$$q(t,z) = e^{j\gamma z|q(t,0)|^2} q(t,0).$$

The nonlinear step is thus $q(t,\epsilon) = e^{j\gamma\epsilon|q(t,0)|^2} q(t,0)$.

ASE noise. This is a simple noise addition

$$q(t, \epsilon^{+}) = q(t, \epsilon^{-}) + n(t),$$

where the noise process $n(t)$ was characterized earlier. The noise can be added in the frequency domain as well.

Question 25. Write an m-function `channel(t, qt, z, nz, betav, gama, sigma2, B)` that outputs a realization of the stochastic process $q(t, z)$ given $q(t, 0)$. Here, betav is the vector of dispersion coefficients.

```
function [qzt, qzf] = channel(t, qt, z, nz, betav, gama, sigma2, B)

f = ...           % get the f vector from t vector
w = ..            % angular frequency
betaw = ...        % beta(omega)
h = exp(j*betaw*dz); % dispersion all-pass filter

qf = ...      % input in frequency

for k=1:nz

   % dispersion step
   qf =  ... % in frequency
   qt = ...  % in time

   % nonlinearity step
   qt =  ...    % in time
   qf = ...     % in frequency

   % noise step
   qf = qf + ...


end
```

Inter-channel interference. One type of distortion in communication is *inter-channel interference*, explained in the class.

Question 26. Simulate the output for the input

$$q(t, 0) = \left(A_1 e^{j\omega_0 t} + A_2 e^{-j\omega_0 t}\right) e^{-\frac{t^2}{2D^2}},$$

where $A_1 = 2A_2 = 2$, $D = 1$, $\omega_0 = 7$ and $z = 1$. Set $\beta_k = 0$ and $n(t, z) = 0$. Plot $\hat{q}(f, 0)$ and $\hat{q}(f, 1)$ in the same graph. Does nonlinearity give rise to interference in the frequency domain (*i.e.*, the two non-overlapping parts in $\hat{q}(f, 0)$ overlap at $z = 1$)? Repeat for $\omega_0 = 16$.
Does nonlinearity give rise to the ISI? Which domain is preferred for communication? □

Modulation, equalization and demodulation. These steps are same as before. The equalization using the SSFM is often called the *digital back-propagation*.

Question 27. We wish to obtain a differential equation whose input is $q(t, \mathcal{L})$ and output is $q(t, 0)$. Introduce change of variable $z = \mathcal{L} - \ell$ in (17) and obtain a differential equation similar to (17) but with different parameters. The back-propagation consists of applying the SSFM to the resulting equation.
Update the equalizer m-function. Write `equalize(t, qt, z, nz, betav, gama)` that outputs $q(t, 0)$ from $q(t, \mathcal{L})$ when noise is zero. □

Detection. The optimal receiver is maximum likelihood applied to the joint conditional PDF $p(\hat{s}^n|s^n)$.

Question 28. Is the channel memoryless according to (15) when $\gamma \neq 0$? Explain. Explain why implementing the optimal receiver is easy in the dispersive channel, but is challenging in the nonlinear dispersive channel.

□

We consider a sub-optimal receiver under the assumption that the channel is memoryless. Here, the ML is applied to the per-symbol conditional PDF $p(\hat{s}_\ell^k|s_\ell^k)$.

Performance evaluation. This step is similar to the corresponding one in the dispersive case, where the BER is computed. Complete the main.m file.

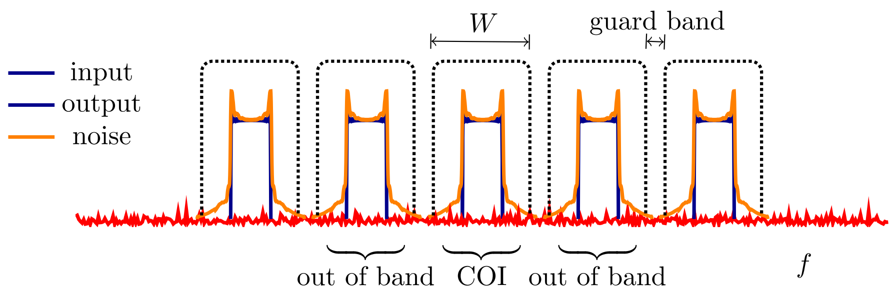
```
% source
b = source(N, p);

% modulation
cnt = constellation(M, R);
s = bit_to_symb(b, cnt);
q0 = mod(t, s, B);
%q0t = A*exp(-t.^2); % to check the code
q0f = ... % input in frequency

% channel parameters, consider normalized equation with no noise
z  = 1; % add other variables
betav = [0, 0, ?] %
gama = 2;
[qzt, qzf] = channel(t, qt, z, nz, betav, gama, sigma2, B);
plot(...) % plot input output in time. Adjust T and N accordingly
plot(...) % plot input output in frequency. Adjust T and N accordingly

[qzte, qzfe] = equalize(t, qt, z, nz, betav, gama);
plot(...) % compare input and output after equalization

% demodulation
shat = demod(t, qzte, B);
```




```
% detection
shat = detector(shat, cnt);
bhat = symb_to_bit(shat, cnt);
```

Verify that BER=0 when $\sigma^2 = 0$ and that the BER is reasonably small when $\sigma^2 \neq 0$. You do not need to plot the BER for several values of SNR and M; you will do this in the next step when WDM is introduced.

□

3.2 Wavelength division multiplexing

You have so far implemented a single-user (or a single-channel) system. We now consider a fiber-optic system with multiple users.

Modulation and multiplexing. We consider a multi-user system with N_u users, N_s symbols per user and bandwidth B. The signal of the user k is centered at the frequency $f = kB_0$ and occupies the frequency band

$$\mathcal{W}_k = \left[kB_0 - \frac{B_0}{2}\ ,\ kB_0 + \frac{B_0}{2}\right],$$

where $B_0 = B/N_u$.
First, the signal $q_k(t, 0)$ of each user k is modulated in the baseband interval $\mathcal{W}_0$. This step is similar to that in the single-user case, *i.e.*,

$$q_k(t,0) = \sum_{\ell=\ell_1}^{\ell_2} s_\ell^k \operatorname{sinc}\big(B_0 t - \ell\big), \tag{18}$$

where k is the user index, ℓ is the symbol index, $(s_\ell^k)_{\ell=\ell_1}^{\ell_2}$ is the sequence of symbols for the user k with zero mean and variance $\mathcal{P}$, and

$$\ell_1 \triangleq -\left\lfloor \frac{N_s}{2} \right\rfloor, \quad \ell_2 \triangleq \left\lceil \frac{N_s}{2} \right\rceil - 1, \tag{19}$$

in which $\lfloor x \rfloor$ and $\lceil x \rceil$ denote, respectively, rounding $x \in \mathbb{R}$ to nearest integers towards minus and plus infinity.

Then, the users' signals are multiplexed by shifting them to the corresponding frequency bands and adding them

$$q(t,0)=\sum_{k=k_1}^{k_2} q_k(t,0)e^{j2\pi kB_0t}, \tag{20}$$

where k_1 and k_2 are defined in terms of N_u similar to (19).

Demultiplexing and demodulation. Let $q(t,z)$ be the received signal when $q(t,0)$ is transmitted. First, demultiplexing is performed, which consists of filtering the received signal on the frequency intervals $\mathcal{W}_k$ for all k and then shifting them to the baseband interval $\mathcal{W}_0$. Denoting the demultiplexed signals by $\tilde{q}_k(t,z)$, we have

$$\tilde{q}_k(t,z)=e^{-j2\pi kB_0t}F_{\mathcal{W}_k}\big(q(t,z)\big), \tag{21}$$

where $F_{\mathcal{W}_k}$ is the filtering operator. Second, the equalization is applied. Finally, the received symbols at the output are obtained by demodulation. This last step, as in the single-user case, is via projection (match filtering):

$$\hat{s}_\ell^k = B\int_{-\infty}^{\infty} E\Big\{\tilde{q}_k(t,z)\Big\}\operatorname{sinc}(B_0t-\ell)\mathrm{d}t,$$

where E is the equalization operator.

Detection. Suppose that the user-of-interest (UOI) is $k=0$. The optimal receiver is again the ML applied to the joint conditional PDF $p(\hat{s}^{N_s}|s^{N_s})$, where $s^{N_s}=(s^0_{\ell_1},\cdots,s^0_{\ell_2})$ is the sequence of symbols of the UOI.
Considering a memoryless approximation

$$p(\hat{s}^{N_s}|s^{N_s})\approx\prod_{\ell=\ell_1}^{\ell_2} p(\hat{s}^0_\ell|s^0_\ell),$$

a sub-optimal receiver is applying the ML to per-symbol conditional PDF $p(\hat{s}^0_0|s^0_0)$.

Question 29. Write an m-function `q=mux(t, Q, B0)` for WDM by implementing (20). Here, Q is a matrix whose rows are users' signals. Write an m-function `Q=demux(t, qz, Nu, B0)` for demultiplexing by implementing (21).

Question 30. Generalize the code for the single-user channel to WDM, by adding multiplexing and demultiplexing functions.

```
% WDM
Nu = 5;
Ns = 5;
B0 = 1; % you may adjust to get meaningful results
```

```
% modulate the users' signals in the baseband
Q= zeros(Nu, N); %
for k=1:Nu
   s(k,:) = bit_to_symb(b, cnt); % generate random symbols from cnt
   Q(k,:) = mod(t, s, B0);
end

% multiplexing
q = mux(t, Q, B0);

% propagation is same as before
qz = ...

% demultiplexing
Q = demux(t, qz, Nu, B0);

% demodulate users' signal
for k=1:Nu
   shat(k,:) = demod(t, Q(k,:), B0);
end

% extract the symbol of interest k=0, l=0
l0, k0= ... % middle symbol of the middle user
s00 = s(l0,k0);
shat00 = shat(l0,k0);
```

You should run this code sufficiently many times and compute `ber(s00, shat00)` or `ber(b00, bhat00)`.

□

3.3 Performance evaluation.

The final results of this section are: (i) the constellation plot; (ii) the plot of the BER or SER as a function of SNR, for an M-QAM constellation with $M = 4, 16$ in the same figure.

You should compare these plots with the ones in the previous section (dispersive channel). The plots are very different. You should explain the origin of the discrepancy and comment in detail.

Remark 2. The system should not operate in a purely-dispersive or purely-nonlinear regime. You may have to adjust the parameters such as B_0, range of a, distance, N_r, etc. □

Remark 3. You need to come up with a procedure to debug your code. For instance, you may first consider the noiseless channel, etc. □

Solitons. A *fundamental soliton* is a signal that keeps its shape in propagation.

Question 31. Simulate (4) with $n(t,z) = 0$ and

$$q(t,0) = A\text{sech}(At), \quad A = 1,2. \tag{22}$$

Plot the absolute value of the input output signals. Is $|q(t,0)| = q(t,z)|$?

□

The shape of *higher-order solitons* can change in distance, sometimes periodically.

Question 32. Repeat the simulation for

$$q(t,0) = A\text{sech}(t). \tag{23}$$

Find values of $A \in (0,6]$ for which the signal shape is spatially periodic. These are Satsuma-Yajima solitons.

□

An important property of solitons is that their collision is elastic.

Question 33. Simulate the output for the input

$$q(t,0) = A_1 e^{j\omega_0 t}\text{sech}\left[A_1(t-t_0)\right] + A_2 e^{\pm j\omega_0 t}\text{sech}\left[A_2(t+t_0)\right].$$

Choose $A_1 = 1$, $A_2 = 2$, $\omega_0 = 1$, t_0 such that the two solitons are well separated in time, and the $\pm$ sign such that the solitons travel towards each other as z is increased. Adjust the ω_0 to set the speed of traveling so that a collision happens in $0 \leqslant z \leqslant 3$.

Choose three suitable values for $z \in (0,3]$, plot $|q(t,z)|$ and show the moment before, during and after the collision.

Explain your observation. □

A Units

The conversion of unit for the loss coefficient α is as follows. Let α denote the loss coefficient for signal in m^{-1}, and $a_{\mathrm{dB}}(z)$ the power loss in the dimensionless unit dB at distance z. Let $P(z)$ denote the signal power at distance z.

The power decays 0.2 dB every 1 km, *i.e.*, $a_{\mathrm{dB}}(1\mathrm{km}) = 0.2$ dB. We have

$$10\log_{10}\left(\frac{P(0)}{P(z)}\right) = a_{\mathrm{dB}}(z), \quad \text{at } z = 1\text{km},$$

or $P(0)/P(z) = 10^{a_{\mathrm{dB}}(z)/10}$ at $z = 1$km. Since $P(0)/P(z) = e^{\alpha z}$, we have

$$\begin{aligned}
\alpha &= \frac{\log(10^{\frac{a_{\mathrm{dB}}(z)}{10}})}{z} \\
&= \frac{\log(10)}{10z} a_{\mathrm{dB}}(z) \\
&= \frac{\log(10)}{10} a_{\mathrm{dB}} \quad \mathrm{km}^{-1} \\
&= 10^{-4}\log(10) a_{\mathrm{dB}} \quad \mathrm{m}^{-1}.
\end{aligned}$$

We write $a_{\mathrm{dB}} = 0.2$ dB/km to mean $a_{\mathrm{dB}}(z = 1\mathrm{km}) = 0.2$ dB. The notation dB/km should not suggest that a_{dB} has unit m^{-1}.

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